

Sneha Sadhwani

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Education

University of California Irvine

Sept 2023 - June 2027

B.S in Computer Engineering, Henry Samueli School of Engineering (GPA: 3.7)

Experience

Software Developer Intern | ClayHR

May 2025 – Aug 2025

- Developed **10+** major features for a Java Spring Boot full stack application and a GPT-4 chatbot
- Backend features used **15+** REST APIs, SQL, and Python Flask. Used Maven to manage dependencies/builds
- Implemented secure user auth, posting and reacting, group chats, interest groups, and chatbot interactions
- Frontend features used JSP, React TypeScript and Tailwind
- Extended project into an Android app that supports **100** users. Used XML, Kotlin, Firebase for database, and Gradle for dependencies/builds
- Collaborated with cross-functional teams using Git and Agile methodologies to deliver reliable features

Undergrad Researcher | UCI Cognitive Robotics Lab

Jan 2025 – June 2025

- Programmed robots' attack and defense maneuvers and a retractable arm using Python and Arduino
- Created **6+** bug reports with clear reproduction steps and identified likely error sources
- Collected and labeled **1,000+** ESP32-CAM images into near/far/none classes, trained custom PyTorch CNN with **89%** accuracy, converted to TensorFlow Lite, deployed on ESP32-CAM for predator robot prey detection

Undergrad Researcher | Air Pollution Monitoring Project ESW @ UCI

Jan 2024 – June 2025

- Programmed Arduino Uno with **4** sensors using I²C, UART, and analog communication protocols
- Developed data logging and calibration routines, processing **1,000+** readings per day for accurate monitoring
- Designed wiring schematics and implemented data transmission protocols for reliable data transfer
- Tested and optimized sensor placements across **8** campus locations to improve data accuracy and coverage

Projects

Matcha: Best Overall Hack VenusHacks 2025 | NextJS, FastAPI, Langchain

May 2025

- Designed and developed a full-stack community app to help UCI students find friends and activities
- Optimized Firestore queries for low-latency, scalable data access supporting **1000+** users under production load
- Integrated Google Gemini API with LangChain to provide personalized activity recommendations for users
- Configured and debugged staging and production CI/CD on Render and Vercel

Full-stack Skin Cancer Detection Website | ReactJS, FastAPI, PyTorch

April 2025

- Trained a CNN model on the **HAM 10000** dataset to recognize skin lesions with **92% accuracy**
- Applied image resizing, normalization, and contrast-enhancement to improve model generalization and accuracy
- Engineered ReactJS/Tailwind frontend and FastAPI backend to enable users to upload skin images for real-time lesion classification in **< 2 seconds**

Autonomous Rover and Remote Controlled Rover | C, IR Sensor

Sept 2023 – March 2024

- Designed and built **2** robots, one autonomous and one remote-controlled, as part of a **6-person** team
- Programmed in C with IR sensors and Pixy camera to enable real-time line tracking and object detection
- Coordinated milestones with Gantt charts and delivered successful live demo showcasing full rover functionality

Skills

- Programming Languages:** Java, Kotlin, Python, C/C++, JavaScript, TypeScript, SQL
- Web & App Development:** React, Tailwind, HTML/CSS, Spring Boot, FastAPI, Firebase, Android Studio
- Tools & Systems:** Git, Gradle, Arduino, PyTorch (Deep Learning), REST APIs, Vercel, Render